

# Ferrite Bead Filters

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**Abstract**—Ferrite beads are commonly used to filter out power rail noise in conjunction with capacitors to filter out power rail noise from sensitive ICs. In this lab, we find that an LC filter can reduce noise by nearly 10x as opposed to a simple through connection or using just a ferrite bead.

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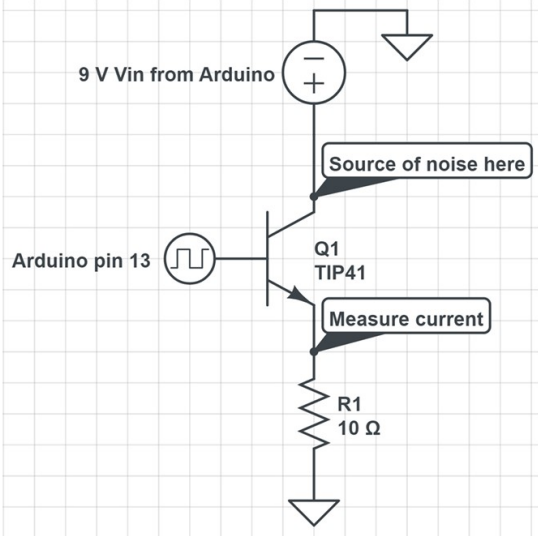


Figure 1. Slammer Circuit

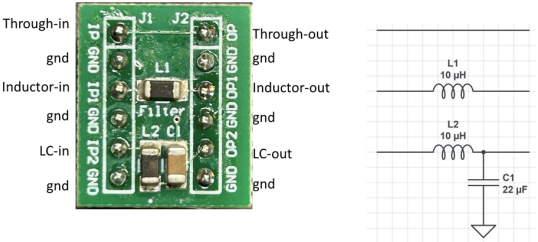


Figure 2. Board Connections

## 1. Introduction

Constructing intricate circuit boards may involve using ICs that are very sensitive to fast switching currents. This becomes a greater concern when the circuit has a significant current draw. In these cases, it may not be enough to use a simple decoupling capacitor. Using a ferrite bead and capacitor together offers a very simple method to construct a dual-pole filter, whereas a single decoupling capacitor is a single-pole filter. This lab will explore the effect of the LC filter on a circuit's power line.

## 2. Objectives

To demonstrate the effect of the ferrite bead, we will create a slammer circuit to simulate a high-current draw circuit (fig. 1) and connect it to a board featuring three different connections. (fig. 2)

1. Through connection
2. Single in-series ferrite bead
3. In-series ferrite bead and parallel capacitor (LC-circuit)

We will then read the voltage at each of the outputs and compare them with each input.

## 3. Plan of Record

### 3.1. What does it mean to work?

Since we are comparing a single-pole with a dual-pole filter, we can expect that the output of the LC filter will contain much less switching noise than a single-pole ferrite filter. Additionally, we know that the position of the ferrite bead plays a significant role in noise reduction. Therefore we must fulfill the following criteria:

1. Generate a significant voltage drop on the through connection line.
2. Generate a smaller voltage drop on the single-pole filter.
3. Generate the smallest voltage drop
4. Observe a difference in noise reduction when the input signal is placed at the output in 2

### 3.2. Risk Reduction

1. The duty cycle of the PWM to the gate pin is kept to 10% to ensure the safety of the 10 ohm resistor.

## 4. Components

1. Arduino (or equivalent) development board.
2. TIP41 x1
3. 10Ω resistor x1
4. Filter breakout board x1

## 5. Arduino Code and Breakdown

```
void setup() {
    // put your setup code here, to run once:
    pinMode(13, OUTPUT);
}

void loop() {
    // put your main code here, to run repeatedly:
    digitalWrite(13, HIGH);
    delay(10);
    digitalWrite(13, LOW);
    delay(100);
}
```

Figure 3. 10% Duty Cycle Generation Code

The code to generate the PWM is very simple. We only need to write high to the PWM pin, then low for 10x longer in order to generate a PWM with a 10% duty cycle.

## 6. Board Construction

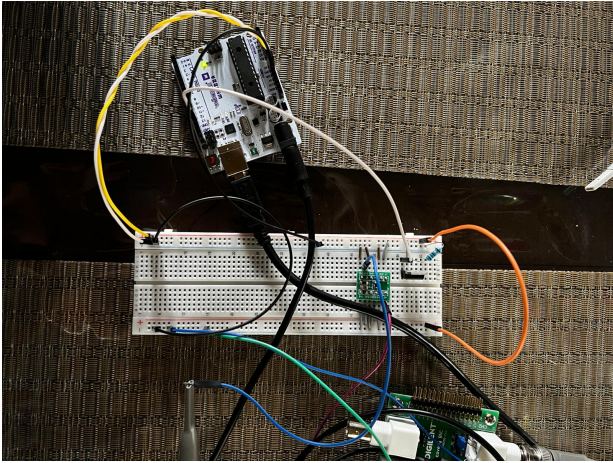


Figure 4. SBB version of fig.1

## 7. Noise Reduction With Normal Connection

The first set of results follows the configuration shown in fig. 2.

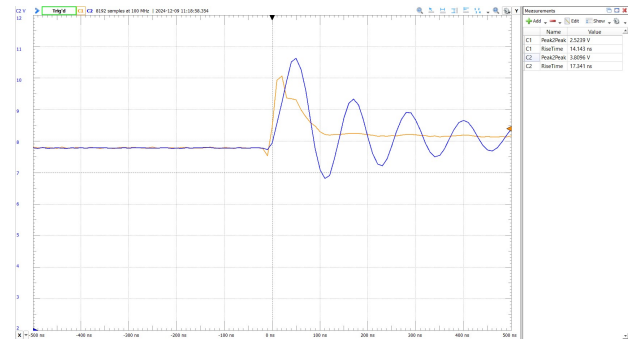


Figure 5. Noise on the through connection (orange) and ferrite bead connection (blue)

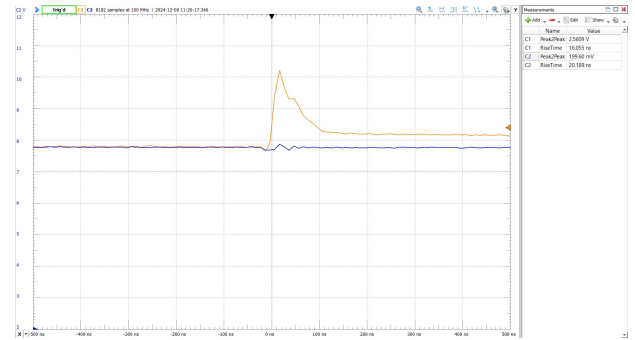


Figure 6. Noise on the through connection (orange) and LC-filter (blue)

From the above figures, we see that the through connection experiences a drop of around 2.5V while the ferrite filter line experiences a drop of around 3.8V, which seems like an improvement. We should experience a much larger voltage drop on the through-line according to the lab guide, however, since we are on the order of nanoseconds, the AD2 may not have the sampling rate necessary to fully capture the voltage drop. Although fig. 5 does not show it, the through-line should experience a drop of around 4V to 4.5V. This means that the ferrite filter provides an improvement in signal integrity when speaking about p-p voltage; the ringing leaves much to be desired.

On the other hand, the LC-circuit provides a nearly 20x noise reduction with no ringing.

## 8. Noise Reduction With Reversed Connection

The second set of results swaps the input and outputs from fig. 2. For easy comparison, we will trigger on the rising edge of the Arduino PWM and show both the noise from the LC-configuration and CL-configuration.

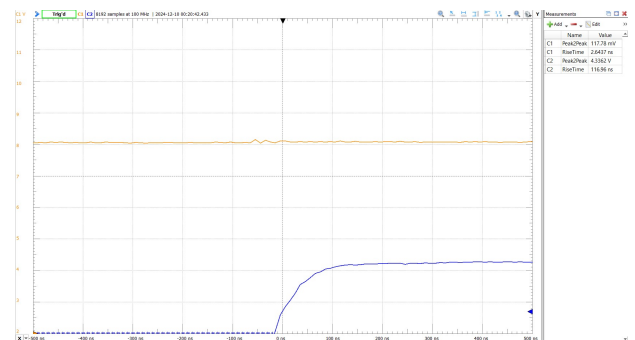
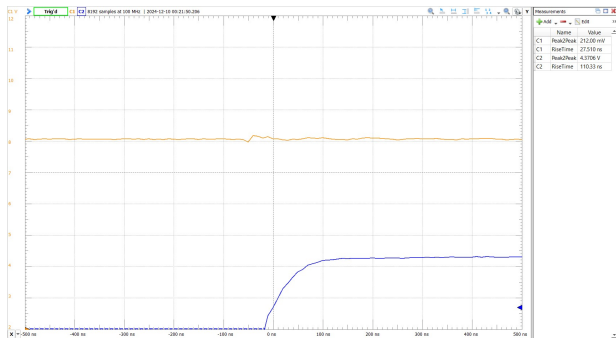


Figure 7. Noise on the line in the LC-configuration (orange) and Arduino PWM (blue)



**Figure 8.** Noise on the line in the CL-configuration (orange) and Arduino PWM (blue)

From the above figures, we see that in the LC-circuit (fig. ??), we average a drop of around 117mV p-p while in the CL-configuration (fig. 8) we get an average drop of around 212mV p-p. In the LC-configuration, we get nearly twice the reduction in noise than in the CL-configuration as we expected.

## 9. Conclusion

Ferrite-capacitor filters (LC-filters) are commonplace in complex boards since they offer an easy way to implement a dual-pole low-pass filter. These filters reduce much more noise than a standard decoupling capacitor filter (i.e. a single-pole filter).

From this lab, we find that the LC-circuit provides a very large reduction in noise compared to a typical through connection and series ferrite filter. However, using a CL-configuration provides slightly less reduction. The following are key takeaways:

1. The configuration of the LC-filter is important to consider when filtering out power-line noise. Using a LC-configuration rather than a CL-configuration provides better signal integrity at no extra cost. Therefore, as best practice, the ferrite bead should be placed closer to the noise source.
2. Ferrite beads on their own are not the best for decoupling noise from power lines. Although in concept, they should provide the same level of filtering as a decoupling-capacitor, in practice they actually leave a significant amount of ringing which is not seen with capacitors. A decoupling capacitor does a much better job of filtering noise as we have seen in previous labs.